PARISHRAM 2.0

CLASS 12TH 2026

MATRIX

MATHS LECTURE-3

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TOPICS TO BE COVERED

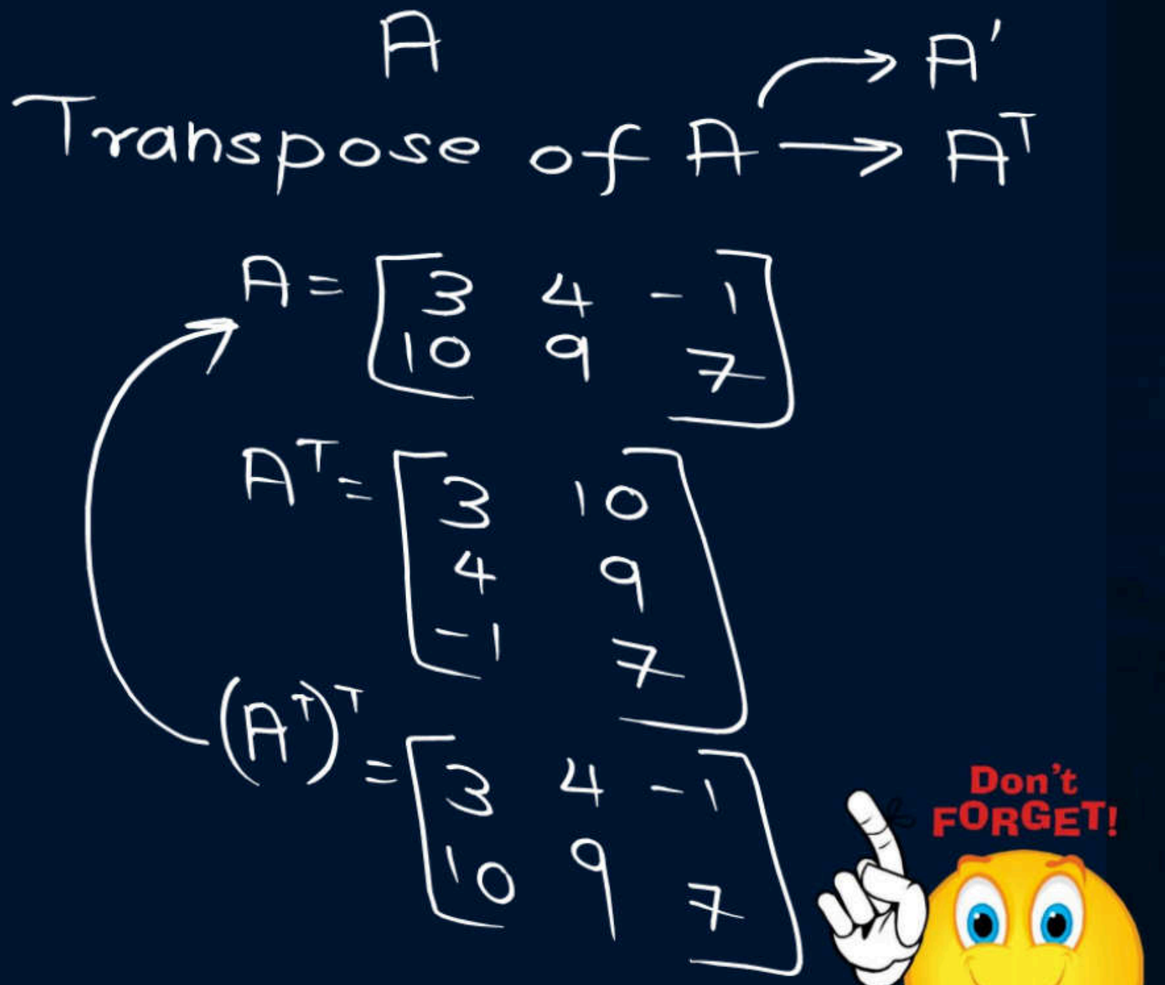
1. Matrix Multiplication
2. Transpose & Symmetric, Skew-Symmetric Questions



Important Formulas

If A′ is the transpose of matrix A, then

1. $(A')' = A$
2. $(A + B)' = A' + B'$
3. $(A - B)' = A' - B'$
4. $(AB)' = B'A'$
5. $(ABC)' = C'\,B'\,A'$





Important Formulas



If A is a square matrix, then

no of rows = no of columns

1. (A + A′) is a symmetric matrix
2. (A – A′) is a skew-symmetric matrix
3. (AA′) and (A′A) are symmetric matrix

Important Formulas

Any square matrix A can be expressed as a sum of symmetric and skew-symmetric matrix:



$$A = \underbrace{\frac{1}{2}(A + A')}_{\text{Symmetric Matrix}} + \underbrace{\frac{1}{2}(A - A')}_{\text{Skew-Symmetric Matrix}}$$

QUESTION [CBSE 2018]

If the matrix $A = \begin{bmatrix} 0 & a & -3 \\ 2 & 0 & -1 \\ b & 1 & 0 \end{bmatrix}$ is skew-symmetric, find the values of 'a' and 'b'.

$$A^T = \begin{bmatrix} 0 & 2 & b \\ a & 0 & 1 \\ -3 & -1 & 0 \end{bmatrix}$$

$$-A^T = \begin{bmatrix} 0 & -2 & -b \\ -a & 0 & -1 \\ 3 & 1 & 0 \end{bmatrix} = A$$

$$\begin{bmatrix} 0 & -2 & -b \\ -a & 0 & -1 \\ 3 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a & -3 \\ 2 & 0 & -1 \\ b & 1 & 0 \end{bmatrix}$$

$A = -A^T$

↳ skew symmetric

$a = -2$

$-b = -3$

$b = 3$

QUESTION

$\cos(-\theta) = +\cos\theta$

$\sin(-\theta) = -\sin\theta$

If $f(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, show that $[f(\theta)]' = f(-\theta)$



$f(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

$[f(\theta)]' = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

LHS

Hence Verified ← RHS $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

RHS

$f(-\theta)$

$f(-\theta) = \begin{bmatrix} \cos(-\theta) & \sin(-\theta) \\ -\sin(-\theta) & \cos(-\theta) \end{bmatrix}$

$= \begin{bmatrix} \cos\theta & -\sin\theta \\ -(-\sin\theta) & \cos\theta \end{bmatrix}$

QUESTION

If $A^T = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$, then find $A^T - B^T$.



$B = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$

$B^T = \begin{bmatrix} -1 & 1 \\ 2 & 2 \\ 1 & 3 \end{bmatrix}$

$A^T - B^T$

$= \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & -2 \end{bmatrix} - \begin{bmatrix} -1 & 1 \\ 2 & 2 \\ 1 & 3 \end{bmatrix}$

$= \begin{bmatrix} 3-(-1) & 4-1 \\ -1-2 & 2-2 \\ 0-1 & -2-3 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ -3 & 0 \\ -1 & -5 \end{bmatrix}$

Conditions For Matrix Multiplication to be possible

$A = \begin{bmatrix} 1 & 2 \\ 3 & 9 \\ 4 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 9 & 4 & -2 \\ 10 & 2 & 3 \end{bmatrix}$

3×2 2×3

↳ point by point

(AB)

Matrix Multiplication

↳ possible (No of coloumns of A = No of rows of B)

$$AB = \begin{bmatrix} 1(9)+2(10) & 1(4)+2(2) & 1(-2)+2(3) \\ 3(9)+9(10) & 3(4)+9(2) & 3(-2)+9(3) \\ 4(9)+0(10) & 4(4)+0(2) & 4(-2)+0(3) \end{bmatrix}$$

$$AB = \begin{bmatrix} 29 & 8 & 4 \\ 117 & 30 & 21 \\ 36 & 16 & -8 \end{bmatrix}$$

QUESTION

Multiply AB

$$A = \begin{bmatrix} 1 & 3 \\ 7 & 2 \\ 4 & 9 \end{bmatrix}, B = \begin{bmatrix} 5 & 9 & 2 \\ 10 & 8 & 7 \end{bmatrix}$$



3×2 ✓ 2×3

$$AB = \begin{bmatrix} 1 & 3 \\ 7 & 2 \\ 4 & 9 \end{bmatrix} \begin{bmatrix} 5 & 9 & 2 \\ 10 & 8 & 7 \end{bmatrix}$$

Order ??

(3×3)

$$= \begin{bmatrix} 1(5)+3(10) & 1(9)+3(8) & 1(2)+3(7) \\ 7(5)+2(10) & 7(9)+2(8) & 7(2)+2(7) \\ 4(5)+9(10) & 4(9)+9(8) & 4(2)+9(7) \end{bmatrix}$$

Comment

QUESTION

If $A = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}$, then find AB, BA. Also show that $AB \neq BA$.

$$A = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & -4 \\ 10 & 3 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1(2)+(-2)(4)+3(2) & 1(3)+(-2)(5)+3(1) \\ -4(2)+2(4)+5(2) & (-4)(3)+2(5)+5(1) \end{bmatrix}$$

$$= \begin{bmatrix} 2-8+6 & 3-10+3 \\ -8+8+10 & -12+10+5 \end{bmatrix}$$

BA → Multiply

AB = BA ??

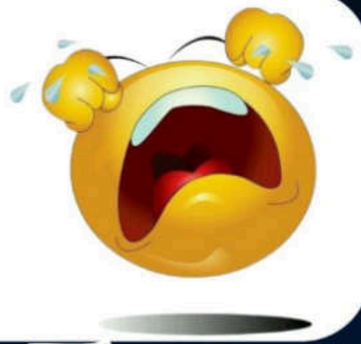
↳ Comment



QUESTION

Multiply XY

$$X = \begin{bmatrix} 4 & 3 & -1 \\ 2 & 4 & 9 \\ 3 & 11 & 0 \end{bmatrix}, \qquad Y = \begin{bmatrix} 7 & 1 & 3 \\ 8 & 9 & 2 \\ 12 & 4 & 2 \end{bmatrix}$$



$$XY = \begin{bmatrix} 4(7)+3(8)+(-1)(12) & 4(1)+3(9)+(-1)(4) & 4(3)+3(2)+(-1)(2) \\ 2(7)+4(8)+9(12) & 2(1)+4(9)+9(4) & 2(3)+4(2)+9(2) \\ 3(7)+11(8)+0(12) & 3(1)+11(9)+0(4) & 3(3)+11(2)+0(2) \end{bmatrix}$$

$$XY = \begin{bmatrix} 40 & 27 & 16 \\ 154 & 74 & 32 \\ 109 & 102 & 31 \end{bmatrix}$$

QUESTION

If $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, find AA^T.



$$A^T = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \quad A^T = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$AA^T = \begin{bmatrix} \cos^2\theta + \sin^2\theta & \cos\theta\sin\theta + (-\sin\theta)\cos\theta \\ \sin\theta\cos\theta - \cos\theta\sin\theta & \sin^2\theta + \cos^2\theta \end{bmatrix}$$

HW

$$A = \begin{bmatrix} 7 & 4 & 3 \\ -2 & 9 & 19 \\ 2 & 4 & 3 \end{bmatrix} \qquad B = \begin{bmatrix} 10 & -7 & 4 \\ 2 & 2 & 11 \\ 0 & 9 & 2 \end{bmatrix}$$

(i) $A+B$

(ii) $A-B$

(iii) $A^T + B^T$

(iv) AB

(v) BA

(vi) $A^T - B^T$

(vii) $A^T \cdot B^T$

Practice

THANK YOU